

EMUQU — Overnight Recovery Measurement from Raw Beat Intervals

Sleep-anchored window selection, score-ranked windows, anti-ratchet baselines and frozen, versioned scores

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Describes Emuqu at source revision e028039 (2026-10-07); scoring version v3.1.oct2026.

Executive Summary

Emuqu is an iOS app that records a night of beat-to-beat (RR) intervals from a chest strap and turns them into one morning number, the recovery score. Most of the steps involved are textbook: RMSSD, the natural log of RMSSD, a rolling personal baseline, z-scores, DFA $\alpha 1$. What is not textbook is everything between the raw beats and that arithmetic. That includes which few minutes of an 8-hour night to measure, how to keep a noisy or short night from moving the baseline, how to assemble one clean series from two imperfect sources, and how to make sure a score shown at 07:00 is still the same score at 19:00 and can be attributed to the algorithm that produced it.

This paper describes those steps precisely enough to re-implement them. The main methods are:

1. **Source assembly.** The strap's internal recording is preferred. It is gap-filled from the Bluetooth stream only inside its own gaps, matched on the phone's wall clock and de-duplicated within 50 ms. An undated strap file is dated by aligning an RR-detected sleep onset to the HealthKit sleep start.
2. **Sleep-anchored window search.** The analysis window is searched only within 30–70% of *measured sleep*, not of the recording. The sleep boundaries are first translated from wall-clock time onto the beat timeline, which stalls during Bluetooth dropouts.
3. **Score-ranked window selection.** Candidate windows that pass artifact, clean-beat and stability gates are ranked by the *same* HRV sub-score function that produces the final score, against the *same* prior-nights baseline. This means the app never picks a window that scores worse than another eligible window under its own scoring rule.
4. **Anti-ratchet baseline.** Nights are admitted to the baseline on structural grounds only, the same way whether they land above or below the mean. Each night has exactly one slot, keyed by wake date. Each night is scored only against earlier nights.
5. **Frozen, versioned scores.** At acceptance the score, its breakdown and its inputs are frozen and stamped with a scoring-version string. Records with no stamp are labelled `unversioned`, not assumed current. Later automatic changes follow fill-only and bounded-upgrade rules.
6. **RR-only sleep staging** by within-night rank normalisation, and a conservative Apple Watch augmentation that overrides Watch stages only on strong evidence.

Section 4 states fourteen specific novelty claims. Section 5 states what has and has not been validated, using the status words of the project's own science register. The time-domain arithmetic is checked on every build against 20 PhysioNet records. The window-selection heuristics, composite weights and sleep staging are not validated against an outcome or against polysomnography.

1. The Problem

1.1 What an overnight RMSSD is supposed to be

The literature on HRV-guided training uses a resting, vagally mediated index, usually $\ln(\text{RMSSD})$, measured under repeatable conditions. Each day's value is compared with a rolling personal baseline, and a change is called meaningful only beyond a "smallest worthwhile change" of about 0.5 SD [2][3][11]. RMSSD itself is defined over successive differences between *adjacent* normal-to-normal intervals [1].

Morning spot readings get repeatability from protocol (same time, same posture, same duration). An overnight recording has no protocol. It contains wake, light sleep, deep sleep, REM, arousals, position changes, strap shifts and, for a Bluetooth stream, dropouts. "The night's RMSSD" is not one number. It depends on which part of the night is measured.

1.2 Sources of error

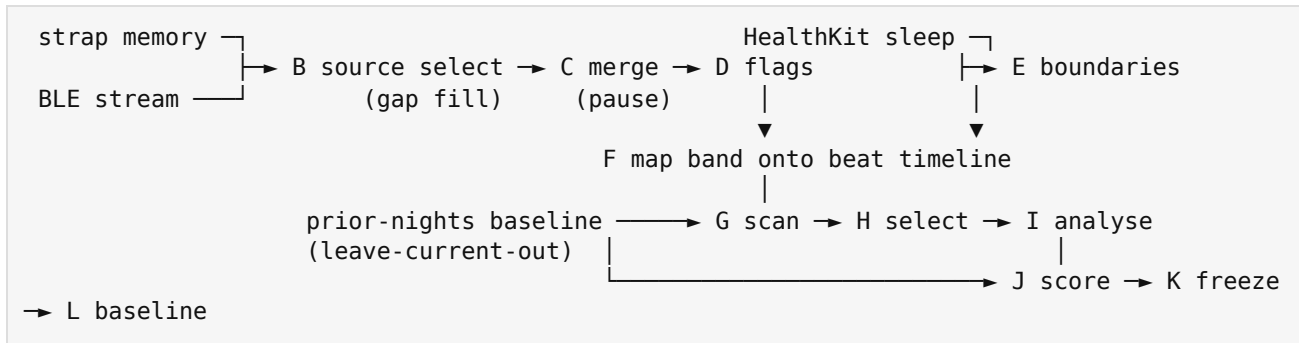
Source	Effect on the reported value
Whole-night averaging	Mixes wake and REM (lower RMSSD) with deep NREM (higher). The value tracks sleep architecture as much as recovery.
Fixed clock window (e.g. "first 4 h", "02:00–03:00")	Lands at a different sleep phase each night when bedtime or sleep onset varies.
Window chosen relative to the recording	Pre-sleep wake time and early removal shift the window. A recording started 90 min before sleep places "30%" in early-night wake.
Bluetooth dropouts	The beat-sum timeline (cumulative RR) stops advancing while wall time continues. Wall-clock sleep boundaries compared directly with beat-sum time land on the wrong beats.
Ectopic beats and missed detections	One compensatory pause can add tens of ms to RMSSD.
Artifact removal by array collapse	Dropping a bad beat and differencing the shortened array creates one spurious large difference per removed beat [1].
Short windows	Short windows have the largest positive RMSSD sampling error. A rule that keeps them only when they are high biases the baseline upward.
Scoring a night against a baseline that contains it	Pulls the z-score toward 0. Kiviniemi et al. compare a day with <i>earlier</i> measurements [6].
Late-arriving sleep data	Apple Watch sleep often reaches HealthKit after the morning score is computed. Without rules, the score silently changes later, or never benefits from the better data.
Algorithm changes	Re-tuning a scoring rule rewrites history unless scores carry a version.

1.3 Why existing approaches are incomplete

The published methods define the statistics (Task Force [1]; $\ln(\text{RMSSD})$ and SWC [2][3]; DFA [4]), the artifact detectors (Kubios-style median-deviation classification [5][10]) and the general idea of HRV sleep staging [13][14][15]. They do not define how to choose an analysis window from an unattended overnight recording, how to keep window choice consistent with a downstream score, how to protect a rolling baseline from selection bias, or how to manage score immutability across devices and late data. Commercial wearables report overnight HRV-derived scores but do not publish their window-selection or baseline-admission algorithms in reproducible detail (Section 6).

2. Architecture Overview

Phase	Operation	Purpose
A. Capture	Stream RR over Bluetooth; strap records internally in parallel; append-only JSONL backup every 60 s	No single failure loses the night
B. Source selection	Prefer internal recording; gap-only composite with stream; date undated files	One continuous, correctly dated series
C. Segment merge	Time-shift resumed segments onto the parent's timeline and concatenate	One window search across a paused night
D. Artifact flags	Whole-series rolling-median classifier	Mark technical, extra, missed and ectopic beats
E. Sleep boundaries	HealthKit $\rightarrow$ RR heart-rate estimate $\rightarrow$ Watch passive-HR estimate $\rightarrow$ recording bounds	Anchor the search to sleep, not to the recording
F. Band mapping	Wall-clock boundaries $\rightarrow$ beat-sum timeline; 30–70% band $\rightarrow$ index range	Correct beats despite dropouts
G. Window scan	Adaptive window, two-pass artifact cap, local ectopic gate, masked RMSSD , $\text{DFA } \alpha 1$	Candidate windows with honest metrics
H. Window selection	Spike filter $\rightarrow$ organized filter $\rightarrow$ rank by HRV sub-score vs prior-nights baseline	Window agrees with the score
I. Analysis	Time, frequency and nonlinear metrics on the chosen window	Reported values
J. Scoring	Tier 1/2/3 composite, adjustments, penalties	0–100 recovery score
K. Freeze	Snapshot sleep/vitals, quality decision, compute, stamp version, archive	Stable, attributable score
L. Baseline update	Structural admission, one slot per night	Unbiased reference for later nights
M. Later events	Bounded automatic rescore; fill-only merges; user edits win	Late data used without silent drift



3. Methods in Detail

Notation: each beat is `RRPoint{t_ms, rr_ms, wallClockMs?, hr?}`. `t_ms` is the beat's time on the recording's beat-sum timeline (cumulative RR). `endMs = t_ms + rr_ms`. `wallClockMs` is the phone's arrival time relative to session start, present on streamed beats only. A valid RR is $300 \leq rr_ms \leq 2000$.

3.1 Capture and source selection

3.1.1 Parallel capture and backup. During an overnight session the app streams RR over Bluetooth and, where the strap supports it (Polar H10), arms the strap's internal recording. Streamed beats are written to an append-only JSONL file: a header file at first write, then one JSON object per beat per line. Only beats beyond the stored count are appended. A flush is due when at least 60 s have passed since the last one, and it is forced on reconnect and lifecycle events. Files use a protection class that stays writable while the phone is locked. A failed append throws, and the beats remain buffered for the next interval.

3.1.2 Strap rescue before clear. Starting a strap recording requires clearing its memory, which is destructive. The start sequence is fixed as `[rescueExisting, clearExisting, beginRecording]`. The rescue step downloads any stored recording that a download ledger says the app does not already hold and hands it to the backup. If that download fails, the start is aborted instead of clearing. A test asserts the invariant "rescue precedes clear" for every strap type.

3.1.3 Source selection. Inputs are the streamed points `S` and the internal points `I` (possibly absent).

1. A source is *valid* if it has ≥ 120 beats.
2. If `I` is invalid: use `S` if valid ("streaming (internal failed)"), else fail.
3. If `I` is valid and `S` is invalid: use `I`.
4. If both are valid, compute `diff = |#I - #S|` and `pct = 100 · diff / max(#I, #S)`. If `#I < #S` and `pct > 5`, go to step 5. Otherwise use `I` (preferred).
5. **Gap detection on `I`.** For consecutive internal beats `prev`, `next`, a gap exists when `next.t_ms - prev.endMs > prev.rr_ms + 2000`. The gap spans `[prev.endMs, next.t_ms]`.
6. **Gap fill.** Select streamed beats whose effective time `wallClockMs ?? t_ms` lies inside any gap (inclusive). Streamed `t_ms` is not used because it lags after Bluetooth packet loss. If there are no gaps, or no streamed beat falls in any gap, the composite fails.

7. **Merge.** Two-pointer merge of `I` with the gap-fill beats, ordered by `I.t_ms` versus the streamed beat's effective time. When an internal beat is emitted, *all* following streamed beats within $|\Delta| < 50$ ms of it are skipped, and vice versa (cluster-aware de-duplication). Emitted streamed beats have `t_ms` rewritten to `wallClockMs`.
8. If the composite fails, use `S` (the larger set), not `I`.

Only gap-interior streamed beats are merged. The phone's arrival clock rarely falls within 50 ms of the strap's own beat times, so merging the whole stream would duplicate most of the night.

3.1.4 Dating an undated strap recording. When a downloaded strap recording has no start time and no live, persisted or uniquely recoverable session claims it:

1. Query HealthKit sleep for the 24 h ending now, passing the recording's beats.
2. If HealthKit reports a sleep start `T_hk`, detect the sleep onset inside the beats (below) as `onsetMs` (0 if none).
3. Recording start = `T_hk - onsetMs`.
4. With no HealthKit sleep, recording start = `now - duration(beats)`.

RR sleep-onset detector (adaptive ceiling):

1. Require > 300 beats.
2. Slide a 120-beat window in 30-beat steps. For each window, keep beats with valid RR. If more than half are valid, record `HR = 60000 / mean(validRR)` at the window's middle beat time.
3. Require > 15 windows.
4. `ceiling = maxHR - 0.5 * (maxHR - minHR)` over this night's windows (midpoint of the night's own HR range).
5. For `j` from 5 to `count - 11`: `before = mean(HR[j-5..j-1])`, `after = mean(HR[j..j+9])`. The first `j` with `before - after > 8` bpm and `after < ceiling` is the onset.
6. Clamp the onset to ≤ 20 min, the plausible sleep-onset latency for a recording started at bedtime [17].

A fixed ceiling such as 65 bpm fails for anyone whose sleeping HR sits above it and has no effect for athletes. The midpoint of the night's own range adapts to each person.

3.2 Pause/resume concatenation

A user may pause an overnight recording (for example, on waking briefly) and resume it. A pause archives a paused session, analysed if it has enough beats and stored raw otherwise. A resume creates a child session linked to the parent. At the next pause or stop, the child's beats are merged with the parent:

1. `offsetMs = child.startDate - parent.startDate`. If `offsetMs ≤ 0` or the value cannot be represented, the merge is refused and the child is analysed alone.
2. Every child beat gets `t_ms += offsetMs` and `wallClockMs += offsetMs`.
3. Merged series = parent beats followed by shifted child beats. Session start = parent start.

The parent already carries its own parent's beats, so a chain of pauses yields the whole night in one series. The pause interval becomes a recording break (Section 3.3.3), so no successive difference is taken across it. Window

selection (Section 3.5) then runs once over the whole night.

3.3 Artifact handling

All artifact handling *drops* beats. Nothing is interpolated.

3.3.1 Whole-series classifier. For each beat, compute the median of a centred window of up to 51 RR values (± 25 beats, including the beat itself, truncated at the ends). With m = median and r = `rr_ms`:

Condition (evaluated in order)	Flag
$r < 300$ or $r > 2000$	technical artifact
$r < 0.5 \cdot m$	extra beat
$0.5 \cdot m \leq r < 0.7 \cdot m$	ectopic
$r > 1.5 \cdot m$	missed beat
$1.2 \cdot m < r \leq 1.5 \cdot m$	ectopic
`	$r - m$
otherwise	clean

The ectopic band is symmetric at 20%. A post-ectopic compensatory pause (20–50% long) is explicitly flagged.

3.3.2 Per-window local ectopic gate. Inside each candidate window, the artifact-free, range-valid RR values are listed in order. Each value is kept if $|r - med| / med \leq 0.20$, where `med` is the median of up to 10 neighbours (5 before and 5 after in that list, **excluding the beat itself**, so a wild value cannot pull its own reference). If the window has ≤ 10 values, every beat is kept. The reported time-domain metrics apply the same gate, so the selector and the report agree.

3.3.3 Masked RMSSD with recording breaks. A *recording break* separates beat `i` from beat `i-1` when either

- `t_i - endMs_{i-1} > rr_{i-1} + 2000`, or
- both have wall clocks and `wall_i - wall_{i-1} > rr_i + rr_{i-1} + 2000`.

RMSSD is the root mean square of `rr_k - rr_{k-1}` over pairs where (a) both beats are kept by every gate, (b) their *original* series indices are consecutive, and (c) beat `k` does not follow a break. When no pair qualifies the value is 0 (selection) or missing (reporting). In reporting only, if the ectopic gate leaves no pair, artifact-clean adjacent pairs are used without it.

3.4 Sleep boundaries and the search band

3.4.1 Boundary sources. On the morning path, boundaries come from, in order:

1. HealthKit sleep for the recording, fetched with the RR series so the HealthKit layer can cross-check it.
2. If HealthKit gives no start: an RR heart-rate estimate. Take 5-min HR windows (≥ 10 beats each) and set `threshold = maxHR - 0.5 \cdot (maxHR - minHR)`. Onset is the first of three consecutive windows below the threshold, clamped to ≤ 20 min. Wake is the window after the last window below the threshold.

3. If no sleep data was obtained at all: an estimate from Apple Watch passive HR samples (same midpoint rule, two consecutive points, ≥ 8 bpm range required).
4. Otherwise: recording start and end.

A negative onset is clamped to 0. HealthKit/HR reconciliation, where used, lets HR move the onset earlier freely. HR can move it later only when HealthKit has no detailed stages, and only by more than 15 min. HR can only extend the wake time.

3.4.2 Wall-clock to beat-timeline mapping. Boundaries arrive as wall-clock offsets from session start. The scan is indexed on `t_ms`, which stalls across dropouts. Mapping: if the first beat has no wall clock (strap internal recording, which has no gaps), the offset is used unchanged. Otherwise, find the first beat with `wallClockMs  $\geq$  firstWall + offset` and return `that.t_ms - first.t_ms`, or the last beat's offset if none qualifies.

- `sleepStart = max(0, map(sleepStartMs))`, or 0 if absent.
- `sleepEnd = min(map(wakeMs), recordingDuration)` if it is later than `sleepStart`, else `recordingDuration`.

3.4.3 Band. `early = sleepStart + 0.30 · (sleepEnd - sleepStart)`, `late = sleepStart + 0.70 · (...)`, clamped to the recording's first and last `t_ms`. The band's index range runs from the first beat with `t_ms  $\geq$  early` to the last beat with `t_ms  $\leq$  late`. A series under 120 beats, or an empty band, yields no window.

Rationale: the early third of sleep is dominated by deep-NREM peaks and the last third by REM and arousals. The middle 40% of *measured sleep* is the most repeatable region from night to night when bedtime, onset and wake vary.

3.5 Window selection

3.5.1 Adaptive geometry. With `B` beats in the band:

- `size = 400` if `B  $\geq$  400`; else `B/2` if `B  $\geq$  240`; else `max(60, [0.6 · B])`.
- `step = max(10, [size/10])`, so 40 beats for a 400-beat window.

Windows are `[e - size, e)` for `e = bandStart + size, bandStart + size + step, ...,  $\leq$  bandEnd`. A 400-beat window is about 6–7 min at 60 bpm.

3.5.2 Per-window evaluation. For each window:

1. `artifactRate = (#flagged) / size`. Reject if above the current pass limit (Section 3.5.3).
2. Valid values = beats that are not flagged and are range-valid. Reject if fewer than 50.
3. Apply the local ectopic gate (Section 3.3.2). Reject if `clean < min(300, max(50, [0.75 · size]))`.
4. `RMSSD` = masked `RMSSD` (Section 3.3.3). `meanRR` and `SDNN` (population, divisor `N`) are taken over clean beats, and `CV = SDNN / meanRR`.
5. `meanHR` = 5–95% trimmed mean of sensor HR values if ≥ 10 are present, plain mean if fewer, else `60000 / meanRR`.
6. DFA α_1 [4] on the clean beats if there are ≥ 64 : integrate, use box sizes 4–16 log-spaced by $2^{1/8}$, apply linear detrend per box, and take the slope of $\log F(n)$ against $\log n$.
7. `relativePosition = (midpoint t_ms - sleepStart) / (sleepEnd - sleepStart)`.

8. LF/HF is **not** computed per window. The field is always empty (see 3.5.4).

3.5.3 Two-pass artifact cap. Scan the band with a 10% artifact cap. Only if that yields **zero valid windows** of any kind, rescan with 15%.

3.5.4 Spike filter and organized filter.

1. *Isolated spike.* Window i (not first or last) is a spike if both neighbours have $\text{RMSSD} > 0$ and $\text{RMSSD}_i \geq 1.5 \cdot \text{RMSSD}_{\{i-1\}}$ and $\text{RMSSD}_i \geq 1.5 \cdot \text{RMSSD}_{\{i+1\}}$. Spikes are removed. If removal leaves nothing, the unfiltered list is used.
2. *Organized.* With α_1 available: $0.75 \leq \alpha_1 \leq 1.0$ **and** $\text{CV} < 0.08$. With α_1 unavailable: $\text{CV} < 0.08$. (The code also contains an alternative rule, " α_1 in range and ($\text{LF}/\text{HF} \leq 1.5$ or $\text{CV} < 8\%$)", for windows that carry an LF/HF value. The production selector never supplies one, so that alternative is dormant and $\text{CV} < 8\%$ is always required. LF/HF is not treated as a sympathovagal measure in any case [7].)

"Organized" names a quiet, stable stretch that makes a good measurement window. It is not a claim about autonomic state.

3.5.5 Ranking by the scoring function. Let `base` be the leave-current-out baseline for this night (Section 3.6.4).

- If `base` exists, each organized window gets $s = \text{Tier1}(\text{rmssd}, \text{meanHR}, \alpha_1, \text{base})$ (Section 3.7.2, with no ANS-balance term and no legacy readiness). Sort by s descending, then RMSSD descending, then relative position descending (later wins).
- If no baseline exists, sort by RMSSD descending, then relative position descending.
- The first window is selected. It is marked *consolidated* if it survived the spike filter and $\text{CV} < 0.08$.

Rationale: the score is not monotone in RMSSD. The z-mapping is flat for $-0.5 \leq z \leq +0.5$ and flat again above $z = +1.5$. The resting-HR and α_1 adjustments (± 10 and $+5/-3/-5$ points) are window-specific. Picking the highest RMSSD can therefore pick a window that scores lower than another eligible window under the app's own rule. A logged example on one night was RMSSD 53 scoring 80 against RMSSD 45 scoring 85. Ranking with the same function and the same baseline makes window choice and score consistent by construction.

3.5.6 Fallbacks and peak capacity.

- If no organized window exists, a *peak-RMSSD fallback* runs over the same band: the same two-pass scan and spike filter, then the window with maximum RMSSD. Its `isOrganized` flag is its own classification, which is false in practice because the same windows have already failed the organized test.
- If even that yields nothing, the whole session is analysed and marked not organized.
- *Peak capacity* is computed independently over the entire recording, not the band. It uses a compact window rule (400 if $B \geq 400$; else $B/3$ if $B \geq 360$; else $\max(60, \lfloor 0.4 \cdot B \rfloor)$), the 15% artifact cap, the spike filter, and no organization filter, and it reports the maximum-RMSSD window. It is shown to the user but is not scored.

3.6 Baseline

3.6.1 Admission (three gates, in order).

1. Session type is overnight.
2. Data quality is not `insufficient` and not `preSleep` (Section 3.8.3).
3. *Structural soundness*: the analysis window lasts $\geq 300\,000$ ms (5 min), **and** the window is organized **or** the session lasts $\geq 10\,800$ s (3 h).

Gate 3 deliberately ignores the direction of the reading. The display path treats a short window differently depending on whether it lands above or below baseline (Section 3.8.3), because a brief clean high reading is still worth showing. For aggregation, that asymmetry is a ratchet. Short windows have the largest positive sampling error. Admitting the high ones and rejecting the low ones raises `lnRmssdMean`, every later night scores lower against it, and the correcting samples are exactly the ones being excluded. Admission is therefore symmetric: a 4-minute window is not baseline-quality whichever side of the mean it falls on.

3.6.2 One slot per night. The night key is the calendar date of the first expected wake after the night's overnight window opens (from the user's sleep schedule). A 23:30 start and its 00:30 continuation share a key. If the key is new, the point is appended. Otherwise the newcomer replaces the incumbent only if:

1. **Morning reading.** If exactly one of the two ended inside the morning window, the morning reading wins.
2. **Consolidation.** If the incumbent is consolidated and the newcomer is not, the newcomer's readiness must exceed the incumbent's by more than 15%. If the newcomer is consolidated and the incumbent is not, the newcomer wins if its readiness is $\geq 90\%$ of the incumbent's.
3. **Organized** (consolidation equal). An organized incumbent beats an unorganized newcomer. An organized newcomer wins if its readiness is $\geq 95\%$ of the incumbent's.
4. **Quality gates.** Reject if the newcomer's artifact % exceeds $\max(1.5 \times \text{incumbent's}, 2.0)$, or its window CV exceeds $1.3 \times$ the incumbent's.
5. **Improvement.** Replace only if readiness improves by more than 5% of the incumbent's.

Missing values default to 0 (readiness), false (flags) and 100% (artifacts). Up to 90 points are stored, sorted by date.

3.6.3 Statistics. Take the 60 most recent stored *nights* (a count, not calendar days). Require ≥ 3 points and ≥ 3 positive RMSSD values.

- `lnMean = mean(ln RMSSD)`.
- `lnSD = max(sampleSD(ln RMSSD) * w, 0.10)`, with $w = 1$ if $n \geq 7$, else $w = \sqrt{7/n}$ (≈ 1.53 at $n = 3$).
- `hrMean, hrSD = max(sampleSD(meanHR), 0.5)`.
- `CV7 = 100 * CV(ln RMSSD of the last 7 points)` if ≥ 7 points (≥ 3 positive), else none.
- `lastDate = date of the most recent point`.

The widening keeps early z-scores near zero until the spread can be trusted, without withholding a score for the first week. The 0.10 floor reflects the documented 5–10% day-to-day noise of overnight RMSSD, so a stable user's trivial -8% night is not scored as a -8 SD event.

3.6.4 Leave-current-out. To score a night, only points whose night key is strictly earlier than the scored night's key are used. Re-scoring an old night therefore never reads later nights. On the live morning path this means every stored night except tonight. The same statistics feed window ranking (Section 3.5.5) and the final score.

3.7 Recovery score (v3.1.oct2026)

3.7.1 z-mapping. $z = (\ln \text{RMSSD} - \ln \text{Mean}) / \ln \text{SD}$, mapped piecewise-linearly through:

z	-3.0	-1.5	-0.75	-0.5	+0.5	+1.5
score	5	25	64	72	72	90

Values below -3 map to 5 and values above +1.5 map to 90. NaN maps to 50. The flat region from -0.5 to +0.5 is the smallest-worthwhile-change deadband [2][3]. The steeper slope below baseline reflects that a drop is actionable while a rise is ambiguous.

3.7.2 Tier-1 HRV sub-score. With a baseline and RMSSD > 0:

```
T1 = map(z)
+ clamp(-5 * (meanHR - hrMean)/hrSD, -10, +10)           // resting HR
+  $\alpha 1$  term: +5 if  $0.75 \leq \alpha 1 \leq 1.0$ ; -5 if  $\alpha 1 > 1.2$ ; -3 if  $\alpha 1 < 0.60$ ; else 0
+ CV7 term: -5 if CV7 < 2%; -3 if CV7 > 12%; else 0
+ ANS term (PNS - SNS): -6 if < -1.5; -3 if < -0.5; +2 if  $\geq +1.5$ ; else 0
- staleness: if d = days(lastDate → referenceDate)  $\geq 7$ : min(20, 5 * (1 + (d-7)/7))
T1 = clamp(T1, 0, 100)
```

Without a baseline: the legacy readiness $(1-10) \times 10$ if present. Otherwise absolute RMSSD bands: $\geq 60 \rightarrow 85$, $\geq 45 \rightarrow 70$, $\geq 30 \rightarrow 55$, $\geq 20 \rightarrow 40$, else 25. With no RMSSD the score is 50.

3.7.3 Tiers. The vitals sub-score is computed first (Section 3.10). The sleep sub-score comes from the sleep analyser when sleep integration is on.

Tier	Inputs present	Composite
3	sleep and vitals	$0.60 \cdot T1 + 0.25 \cdot \text{Sleep} + 0.15 \cdot \text{Vitals}$; comeback mode $0.80/0.20/0.00$
2	sleep only	$0.70 \cdot T1 + 0.30 \cdot \text{Sleep}$; if $z < -1$ and Sleep < 50: $0.85/0.15$ (double-penalty dampening)
1	neither	T1

Comeback mode is active for 21 days from the date the user sets it. In Tier 3 the vitals factor is still shown, with weight 0. Tier 2 dampening keeps a bad-sleep night from being penalised twice, once through the HRV drop it usually causes and once through the sleep factor.

3.7.4 Final composite.

1. Missing sleep: in Tier 1, if sleep integration is on, the user has opted into the penalty and there is no sleep data, subtract 10.
2. Clamp to 0–100.
3. SpO₂: if a reading exists and is < 95%, subtract a flat 10 and floor at 0. This is a conservative product rule, not a clinical threshold.

Every deduction is listed in the breakdown.

3.7.5 Training load is excluded. Training-load metrics (acute:chronic ratio, monotony, strain) are accepted by the scoring API but not read. Day-to-day ACWR has been shown to carry little signal beyond its acute term [8]. A separate training-readiness value is frozen alongside the score and is described in another paper.

3.8 Score freezing

3.8.1 Snapshot. At acceptance, sleep (for the whole overnight window of the night the session belongs to) and vitals are fetched together with a 10-s timeout. On timeout, scoring proceeds without them. In the vitals snapshot, the **strap's analysis-window mean HR replaces HealthKit's daytime resting HR**, so the vitals HR is compared like-for-like with the baseline's `hrMean`, which is built from the same nocturnal quantity.

3.8.2 Inputs. RMSSD, meanHR and α_1 from the selected window. ANS balance = PNS index – SNS index. Baseline = leave-current-out stats. **Reference date = session end**, not the wall clock, so the staleness term is identical whenever the score is recomputed.

3.8.3 Quality decision. Let `base = exp(lnMean)` (0 if no baseline), `dur` = session length and `win` = analysis-window length (session length if there is no window). The rules are evaluated in order:

1. If (`win < 5 min` or (`not organized` and `dur < 3 h`)) **and** `RMSSD < base`: quality `insufficient`, use the baseline.
2. If sleep data exists and the recording does not overlap it: quality `preSleep`, use the baseline. This applies regardless of RMSSD direction.
3. If `RMSSD < base` and `dur < 3 h`: quality `insufficient`, use the baseline.
4. Otherwise quality `good`. A long night below baseline is a real bad night.

Rules 1 and 3 are asymmetric: a short reading is replaced only when it is low. This affects *display*. Baseline admission (Section 3.6.1) is separately symmetric.

3.8.4 Unusable-reading blend. "Use the baseline" means scoring with `RMSSD = exp(lnMean)`, `meanHR = hrMean`, no α_1 and no readiness, so T1 lands near 72 less any CV or staleness deduction. If the user has answered the subjective readiness prompt ($p \in [0, 1]$, clamped), the HRV factor becomes $0.70 \cdot T1 + 0.30 \cdot (100 \cdot p)$. The composite is rebuilt through the same tail as step 3.7.4, so the missing-sleep and SpO₂ deductions still apply.

3.8.5 What is stored. `recoveryScore` (composite / 10), `scoreBreakdown` (composite, tier, factors, penalties, SpO₂ flag and `scoringVersion = "v3.1.oct2026"`) and the frozen training readiness. A breakdown decoded without a version field is labelled **unversioned**, never assumed to be the current version, because the archive cannot tell which earlier algorithm produced it.

3.8.6 Mutation rules.

- *User edits win.* Bulk reanalysis skips sessions whose analysis window the user picked by hand. Automatic sleep refresh never overwrites a sleep snapshot the user has edited. On a second device, where the cloud copy arrives without a snapshot, it may *fill* the snapshot but keeps the user's boundaries and does not rescore.
- *Automatic paths fill, not overwrite.* Pulled-session sleep boundaries are set only where they are missing. An imported duplicate adopts a recovery score only when the existing record has none.

- *Bounded automatic upgrade.* When HealthKit sleep changes, only the most recent overnight session that ended within the last 18 h and has already been scored is considered. Runs are serialised: overlapping triggers collapse into one trailing pass. The fresh sleep must plausibly overlap the recording. The snapshot is updated if this is the first snapshot, the end moved later by more than 60 s, total minutes increased, the onset moved by ≥ 15 min, or the boundary source upgraded from an estimate (HR, Watch HR or recording bounds) to HealthKit or HR-validated HealthKit. A **rescore** follows only for a first snapshot, a total change of ≥ 20 min, an onset move of ≥ 15 min, or a source upgrade. The scoring boundaries are updated with the snapshot so the rescore uses the new window.
- *Cross-device.* The score, breakdown (with version) and frozen readiness travel inside one encrypted session payload. The newer copy (by whole-second modification stamp) replaces the older as a unit, after which only locally held HealthKit readings are restored.

3.9 RR-only sleep staging

3.9.1 Epochs and features. Between sleep start and end, cut 5-min epochs on the beat timeline. An epoch needs ≥ 10 beats and ≥ 8 range-valid RRs, otherwise it is dropped. Per epoch: HR = 60000 / mean RR, RMSSD, population SDNN, CV = SDNN / mean RR, $\alpha 1$ (if ≥ 64 beats), and LF/HF and HF power from the shared spectral routine over spline-resampled beats.

3.9.2 Within-night rank normalisation. Each feature is replaced by its fractional rank across the night's epochs, from 0 (lowest) to 1 (highest), with ties given the mean rank. Missing values are filled before ranking ($\alpha 1 \rightarrow$ the night's median $\alpha 1$ or 0.85; LF/HF $\rightarrow$ 2.0; HF $\rightarrow$ 0), so a night with no $\alpha 1$ gives every epoch 0.5. Ranking makes the classifier scale-free: it never compares a feature with an absolute threshold, so it adapts to each person and each sensor.

3.9.3 Scores. f = night fraction of the epoch midpoint. Temporal priors: deep $T_d = 1.0$ if $f < 0.4$, 0.4 if $f < 0.6$, else 0.15 . REM $T_r = 1.0$ if $f > 0.6$, 0.25 if $f > 0.35$, else 0 . Writing r_x for the rank of feature x :

Variant (by availability across the night)	Deep score weights on (1- r_{hr} , r_{rmssd} , 1- r_{cv} , r_{hf} , 1- r_{lfhf} , 1- $r_{\alpha 1}$, T_d)	REM score weights on (1- r_{rmssd} , r_{cv} , r_{lfhf} , 1- r_{hf} , $r_{\alpha 1}$, T_r)
LF/HF and $\alpha 1$	.12, .10, .08, .15, .18, .22, .15	.10, .10, .24, .12, .26, .18
LF/HF only	.15, .13, .10, .18, .24, —, .20	.12, .12, .30, .18, —, .28
$\alpha 1$ only	.18, .18, .12, —, —, .32, .20	.16, .16, —, —, .38, .30
time-domain only	.25, .25, .15, —, —, —, .35	.32, .43, —, —, —, .25

Awake score: with LF/HF, $.30 \cdot r_{hr} + .25 \cdot (1 - r_{rmssd}) + .15 \cdot r_{cv} + .20 \cdot r_{lfhf} + .10 \cdot (1 - r_{hf})$. Without, $.35 \cdot r_{hr} + .30 \cdot (1 - r_{rmssd}) + .20 \cdot r_{cv} + .15 \cdot r_{\alpha 1}$. The REM score is forced to 0 in the first 60 min after sleep start.

3.9.4 Decision and smoothing. Awake if $awake > 0.80$. Otherwise deep if $deep > 0.60$ and $deep \geq rem$. Otherwise REM if $rem > 0.63$ and $rem > deep$. Otherwise core. Then, left to right, any non-awake epoch whose previous and next epochs agree on a different stage takes that stage. Awake epochs are never smoothed away.

3.9.5 Intervals. Consecutive same-stage epochs form an interval. A dropped epoch (a gap) also ends the interval, so a strap dropout is never counted as sleep.

3.9.6 Watch augmentation. When Apple Watch stages exist, the Watch is the anchor. Each RR epoch takes the Watch stage covering most of it. Epochs the Watch does not cover are left alone. Overrides:

Watch stage	Override when	To
core / unspecified	deep > 0.68 and deep > rem	deep
core / unspecified	LF/HF present, rem > 0.63, rem > deep	REM
awake	LF/HF present, rem > 0.68, awake < 0.80	REM
deep	LF/HF present, rem > 0.72, deep < 0.40	REM
REM	deep > 0.72, rem < 0.40	deep

Only the part of each Watch interval that carries the overridden stage, inside the overridden epoch, is repainted and tagged as HRV-derived. All other Watch timing is kept and no smoothing touches Watch stages. REM overrides need LF/HF because time-domain features separate REM from N2 poorly [15].

3.10 Vitals sub-score

The vitals sub-score is the mean of whichever sub-scores exist. A missing input is dropped, not penalised. Every rule penalises in one direction only.

Input	Sub-score
Sleep HR (strap window mean HR, Section 3.8.1)	$\max(0, 100 - 10 \cdot \max(0, (HR - \text{hrMean})/\text{hrSD}))$. Requires a baseline. At or below baseline scores 100.
Respiratory rate with a 7-night baseline	$\max(0, 100 - 15 \cdot \max(0, (\text{rate} - \text{baseline}) - 1))$. Below baseline scores 100.
Respiratory rate without a baseline	100 within 12–18 br/min; above 18: $100 - 15 \cdot (\text{rate} - 18)$, floor 0; below 12: 90
Wrist temperature (deviation from the user's own baseline; dropped if there is no baseline)	$\leq +0.3^\circ\text{C} \rightarrow 100$; $\leq +0.5 \rightarrow 75$; $\leq +1.0 \rightarrow 50$; above $\rightarrow 25$. Cooler than baseline scores 100.

With no inputs, vitals is absent and the tier falls back. SpO₂ is not a factor (Section 3.7.4).

4. What Is New

The author is not aware of a prior publication of any of the following as described. Each statement refers to the exact rules in Section 3.

- **N1.** Building an overnight RR series by preferring a chest strap's internal recording and adding Bluetooth-streamed beats only when the internal file is more than 5% shorter. Streamed beats are added only inside internal gaps (next start – previous end > previous RR + 2000 ms), are matched on the phone's arrival clock

rather than their own drifting beat-sum time, and are de-duplicated cluster-wise within 50 ms. If no gap can be filled, the stream is used rather than the shorter internal file.

- **N2.** Dating a chest-strap recording that carries no start time by subtracting an RR-detected sleep onset from the HealthKit sleep start. The onset is the first sustained drop of more than 8 bpm (5-window versus 10-window means over 120-beat windows) into the lower half of that night's own HR range, clamped to 20 min.
- **N3.** Placing the analysis-window search band at 30–70% of *measured sleep*, after translating wall-clock sleep boundaries onto the recording's beat-sum timeline through each beat's arrival timestamp, so the band lands on the correct beats across Bluetooth dropouts.
- **N4.** A candidate-window scan whose size adapts to the beats available in the band (400; else half the band if ≥ 240 ; else 60% with a floor of 60) with a step of $\max(10, \text{size}/10)$, a 10% artifact cap relaxed to 15% only when the strict pass yields no window at all, and a clean-beat floor of $\min(300, \max(50, 0.75 \times \text{size}))$.
- **N5.** Using one masked RMSSD estimator for both window selection and reporting. It takes differences only between beats that were adjacent in the original series, survived both a whole-series classifier and a self-excluding 10-neighbour 20% median gate, and are not separated by a recording break defined on both the beat timeline and the arrival clock.
- **N6.** Selecting among windows that are not isolated spikes ($\geq 150\%$ of both neighbours' RMSSD, with the unfiltered list kept if filtering removes everything) and are "organized" (DFA α_1 in 0.75–1.0 and HR coefficient of variation $< 8\%$, or the CV test alone when α_1 is unavailable).
- **N7.** Ranking candidate analysis windows by the same HRV sub-score function that produces the final recovery score: z-mapping with a flat deadband plus resting-HR and α_1 adjustments. The ranking is computed against the same leave-current-out personal baseline, ties are broken by RMSSD and then by later position, and the method falls back to RMSSD ranking only when no baseline exists.
- **N8.** Reporting a peak-RMSSD fallback window, flagged not organized, when no organized window exists, alongside an independently computed whole-recording "peak capacity" that is displayed but never scored.
- **N9.** Admitting nights to a rolling $\ln(\text{RMSSD})$ baseline by a direction-independent structural test (window ≥ 5 min, and organized or session ≥ 3 h), while the display-side quality rule remains direction-dependent, specifically to prevent short high readings from ratcheting the baseline upward.
- **N10.** Holding one baseline slot per night keyed by the wake date of the user's sleep schedule, with an ordered replacement rule: morning reading, then consolidation (15% / 90%), then organized (95%), then artifact and CV quality ceilings, then a 5% improvement.
- **N11.** Computing baseline statistics over the last 60 stored nights (by count), widening the $\ln(\text{RMSSD})$ SD by $\sqrt{(7/n)}$ below seven nights with a 0.10 floor, and scoring each night only against nights with an earlier night key, with the same statistics used for window ranking and for the score.
- **N12.** Freezing a recovery score at acceptance with: the strap's nocturnal window HR substituted for daytime resting HR in the vitals snapshot; a reference date equal to session end; a baseline-substitution rule for unusable readings that is triggered asymmetrically for short readings, with a 70/30 baseline/subjective blend routed through the same penalty tail; and a stored algorithm-version string, with records lacking one labelled "unversioned" rather than current.
- **N13.** Bounding automatic post-freeze changes. Only the most recent overnight session ended within 18 h is eligible. Rescoring happens only on a first snapshot, a ≥ 20 min total change, a ≥ 15 min onset move, or a boundary-source upgrade from an estimate to Watch-based data. Scoring boundaries move with the display snapshot. User-edited boundaries and manual windows are never overwritten. Score, breakdown and version are merged across devices as one unit.

- **N14.** RR-only sleep staging on 5-min epochs using within-night fractional ranks with tie-averaging and constant fills, four weight sets chosen by which feature families exist that night, time-of-night priors with a 60-min REM lockout, single-epoch smoothing that never removes wake, and gap-terminated intervals. This is combined with a Watch augmentation that repaints only the overridden stage inside overridden epochs of the Watch's own intervals and requires spectral data for any REM override.

Not claimed (prior art). RMSSD, SDNN and pNN50 definitions and the adjacency requirement [1]. $\ln(\text{RMSSD})$ baselines, z-scores and the 0.5 SD smallest worthwhile change [2][3][11]. Comparison with earlier days [6]. DFA and its box-size ranges [4]. Median-deviation artifact and ectopic classification and the 20% threshold [5][10][12][16]. The concept of HRV-based sleep staging and the direction of α_1 and LF/HF changes across stages [13][14][15]. Weighted composite readiness scores in general. The criticism of ACWR [8]. Piecewise-linear score mappings in general.

5. Accepted Tradeoffs and Limitations

Validation status (exact status words from `Tools/science_register/register.json`, scoring version v3.1.oct2026):

Register entry	Status	What it covers here
hrv-time-domain-arithmetic	validated	RMSSD, SDNN and pNN50 arithmetic. Checked every build against 6,000 intervals from 20 PhysioNet nsr2db records [9] versus an independent Task Force implementation. This covers the arithmetic, not the strap's beat detection.
lnrmssd-swc-band	validated	Use of the 0.5 SD smallest worthwhile change on rolling $\ln(\text{RMSSD})$, the published method
resting-dfa-a1-reference-band	awaiting-validation	The 0.75–1.0 α_1 band: the +5 bonus and the organized-window test
lfhf-window-filter	awaiting-validation	Dormant in production (no per-window LF/HF is supplied)
pns-sns-gap-readiness	awaiting-validation	The ANS-balance adjustment
composite-weights-60-25-15	awaiting-validation	Tier weights, RHR and CV adjustments
rmssd-absolute-fallback-bands	awaiting-validation	No-baseline RMSSD bands
baseline-staleness-penalty	awaiting-validation	Staleness deduction
sleep-score-six-factor	awaiting-validation	Sleep sub-score
hrv-sleep-staging	awaiting-validation	Section 3.9. Agreement with polysomnography is unmeasured for this implementation; its accuracy is unknown.

Register entry	Status	What it covers here
spo2-flat-penalty	awaiting-validation	SpO ₂ deduction

The window-selection method (Sections 3.4–3.5), the baseline admission and replacement rules (Section 3.6) and the freeze rules (Section 3.8) are not separate register entries. They have not been validated against any outcome. Their correctness is defined by internal consistency (for example, the window chosen is the one the score prefers) and is enforced by unit tests, not by external evidence.

Tradeoffs and known limitations.

- *A short window represents the night.* About 6–7 min of the middle of sleep stands for the whole night. This trades coverage for repeatability.
- *Selection maximises the score.* Ranking by the scoring function favours the best-scoring eligible window. Nightly scores are therefore optimistic compared with a random eligible window. This is consistent across nights but not neutral.
- *Ranking omits ANS balance.* Window ranking does not include the ANS-balance term, which the final score does include, so the two can differ by up to –6/+2 points. Ranking also evaluates staleness at the current time rather than at session end, but staleness is the same for every window and so does not change the order.
- *Organized depends on unvalidated thresholds.* The $\alpha 1$ band and $CV < 8\%$ are app conventions. A night with no organized window gets a fallback window.
- *The baseline is display-asymmetric but admission-symmetric.* A short high reading is shown to the user but not admitted to the baseline.
- *Estimated sleep boundaries are biased late.* HR-estimated onsets can still be early or late within the 20-min clamp. The search band shifts with them.
- *Sleep-staging RMSSD is not masked.* Per-epoch RMSSD in the sleep classifier differences the collapsed list of valid beats and does not use the masked estimator of Section 3.3.3.
- *Population SDNN in selection.* Window CV and SDNN in selection use the population SD. The reported SDNN uses the sample SD.
- *The app is not a medical device.* No output is a diagnosis. The SpO₂ rule is a conservative product rule.
- *Single-author development and testing.* There is no independent replication.

6. Comparison to Existing Approaches

Approach	Window choice	Baseline	Artifact handling	Score versioning
Task Force 1996 [1]	Short-term (5 min) or 24 h recordings; no overnight window-selection rule	Not specified	Requires NN intervals; method not prescribed	n/a
Plews/Buchheit ln(RMSSD) monitoring [2][3][11]	Standardised morning spot reading	Rolling ln(RMSSD) mean, SWC 0.5 SD	Protocol-based	n/a
Kiviniemi 2007 [6]	Morning measurement	Earlier days only	Not the focus	n/a

Approach	Window choice	Baseline	Artifact handling	Score versioning
Kubios HRV [10] with Lipponen-Tarvainen correction [5]	User-chosen sample or whole recording	None (analysis tool)	Classifies and <i>corrects</i> (interpolates) beats	n/a
HRV sleep-staging literature [13][14] [15]	Epoch-based staging	Not applicable	Varies	n/a
Commercial wearables (e.g. WHOOP, Oura, Garmin, Polar)	Publicly described as using HRV measured during sleep; exact window-selection rules not publicly documented	Not publicly documented in reproducible detail	Not publicly documented	Not publicly documented
Emuqu	30–70% of measured sleep on the beat timeline; adaptive windows ranked by the scoring function	Structural admission, one slot per night, 60 nights, $\sqrt{(7/n)}$ widening, leave-current-out	Drop-only; whole-series classifier + self-excluding local gate; break-aware masked RMSSD	Version string on every score; unversioned sentinel

7. Provenance and Dates

Author and sole committer: Chris Sharp chrissharp80@gmail.com.

Repositories. Private development repository `chrissharp80/emuqu-dev`: first commit 2026-08-16 (squashed from earlier work). Public repository `github.com/chrissharp80/emuqu` (PolyForm Strict 1.0.0): first commit 2026-09-08.

Earlier product name "Flow Recovery" (the author's own records): app-generated `Flow_Recovery_<date>.pdf` session reports in the author's Google Drive dated 2026-02-15, 2026-02-17 and 2026-03-03; a "Flow Recovery - Bluetooth" screen recording dated 2026-03-23; and a "flow-recovery-audit.md" dated 2026-05-10.

First appearance in emuqu-dev history (per file):

File	First added
BaselineTracker, HRVSleepStageClassifier, WindowSelection (+Evaluation, +Filters), DataSourceSelector, SessionAcceptanceService	2026-08-16
WindowSelection+Scoring	2026-08-22
HRVSleepStageClassifier+Watch	2026-08-25
SleepRefreshPolicy	2026-09-02
ScoringVersion	2026-09-03

First appearance of specific mechanisms in the public repository:

Mechanism	Public commit date
Score-ranked windows, wall-clock → beat-timeline mapping, structural baseline admission, $\sqrt{(7/n)}$ SD widening, unversioned sentinel, strap-nocturnal-HR substitution, adaptive onset ceiling, HealthKit-aligned dating, rescue-before-clear, SWC deadband bands, rank-normalised staging	2026-09-08 (initial public commit)
Leave-current-out baseline; gap-interior-only stream merge; Watch-interval repainting	2026-10-03
Recording-break rule and masked RMSSD in selection	2026-10-04

Scoring versions: v2.may2026 (public 2026-09-08) → v3.oct2026 (2026-10-03) → v3.1.oct2026 (2026-10-04).

Validation data: time-domain arithmetic checked against 20 PhysioNet nsr2db records (`docs/hrv-reference-data.md`).

This paper's publication date: 2026-10-08.

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Appendix A: Parameter Table

Name	Value	Unit	Role
minimumValidBeats	120	beats	Source validity; minimum series for band search
compositeThresholdPercent	5	%	Internal deficit that triggers a composite
gap slack	2000	ms	Gap / recording-break tolerance beyond one RR
duplicateToleranceMs	50	ms	Cross-source duplicate window
backup interval	60	s	Append-only JSONL flush
onset minimum beats	> 300	beats	Onset detector input
onset window / step	120 / 30	beats	Onset HR windows
onset minimum windows	> 15	windows	Onset detector
onset drop	> 8	bpm	Before (5-window) – after (10-window) mean
ceiling fraction	0.5	—	Midpoint of the night's HR range
max onset latency	20	min	Clamp on HR-estimated onset
boundary disagreement	15	min	HR vs HealthKit reconciliation
RR valid range	300–2000	ms	Physiological gate
rolling median half-window	25	beats	Whole-series classifier (51-value centred)

Name	Value	Unit	Role
extra / missed thresholds	$0.5\times / 1.5\times$	median	Classifier
ectopic threshold	0.20	fraction	Classifier and local gate
local median neighbours	10	beats	Self-excluding local gate
band	0.30–0.70	sleep fraction	Search band
beatsPerWindow	400	beats	Target window
adaptive window	$B/2$ if $B \geq 240$; else $\max(60, 0.6B)$	beats	Short bands
step	$\max(10, \text{size}/10)$	beats	Scan step
artifact cap strict / fallback	0.10 / 0.15	fraction	Two-pass scan
minimum valid RRs	50	beats	Per window
minCleanBeats	$\min(300, \max(50, 0.75 \cdot \text{size}))$	beats	Per window
DFA minimum	64	beats	$\alpha 1$ computed
$\alpha 1$ boxes	4–16 (ratio $2^{1/8}$)	beats	DFA $\alpha 1$
isolatedSpikeRatio	1.50	$\times$	Spike filter
organized $\alpha 1$	0.75–1.0	—	Organized test
unstable CV	0.08	fraction	Organized and consolidated tests
maxOrganizedLfHf	1.5	—	Dormant alternative
peak-capacity window	400; $B/3$ if $B \geq 360$; else $\max(60, 0.4B)$	beats	Whole-recording scan
reliable window	300 000	ms	Baseline admission; quality rule
overnight session	10 800	s	Baseline admission; quality rule
stored points	90	nights	Baseline storage
stats window	60	nights	Baseline statistics
minimum nights	3	nights	Baseline exists
widening	$\sqrt{(7/n)}$, $n < 7$	—	SD widening
lnRmssdSDFloor	0.10	ln units	SD floor
meanHRSDFloor	0.5	bpm	HR SD floor
replacement fractions	0.15 / 0.90 / 0.95 / 0.05	—	Night-slot replacement
artifact ceiling	$\max(1.5\times, 2.0)$	%	Night-slot replacement
CV reject multiplier	1.3	$\times$	Night-slot replacement
z bands	(−3,5) (−1.5,25) (−0.75,64) (−0.5,72) (0.5,72) (1.5,90)	—	z-mapping
RHR multiplier / clamp	−5 / ± 10	points per SD	Tier-1 adjustment
$\alpha 1$ bonus / penalties	+5 / −5 (>1.2) / −3 (<0.60)	points	Tier-1 adjustment

Name	Value	Unit	Role
CV7 thresholds / penalties	$<2 \rightarrow -5; >12 \rightarrow -3$	% / points	Tier-1 adjustment
ANS thresholds / points	$<-1.5 \rightarrow -6; <-0.5 \rightarrow -3; \geq 1.5 \rightarrow +2$	— / points	Tier-1 adjustment
staleness	≥ 7 d: $\min(20, 5 \cdot (1 + (d-7)/7))$	points	Tier-1 deduction
absolute bands	60/45/30/20 $\rightarrow$ 85/70/55/40; else 25; none 50	ms / points	No-baseline fallback
Tier 3 weights	0.60/0.25/0.15	—	Composite
Comeback weights / duration	0.80/0.20/0.00 / 21	— / days	Composite
Tier 2 weights / dampened	0.70/0.30 / 0.85/0.15	—	Composite
dampening trigger	$z < -1$ and Sleep < 50	—	Tier 2
missing-sleep penalty	10	points	Tier 1
SpO ₂ threshold / penalty	95 / 10	% / points	Post-composite
perceived blend	0.70 / 0.30	—	Unusable reading
snapshot timeout	10	s	Freeze
auto-refresh horizon	18	h	Eligible session
rescore total delta	20	min	Rescore gate
onset move	15	min	Update and rescore gate
end-later tolerance	60	s	Update gate
staging epoch	5	min	Sleep staging
epoch minimum points / valid	10 / 8	beats	Sleep staging
minimum epochs	6	epochs	Sleep staging
REM lockout	60	min	Sleep staging
thresholds awake / deep / REM	0.80 / 0.60 / 0.63	—	Sleep staging
deep prior	$<0.4 \rightarrow 1.0; <0.6 \rightarrow 0.4; \text{else } 0.15$	—	Sleep staging
REM prior	$>0.6 \rightarrow 1.0; >0.35 \rightarrow 0.25; \text{else } 0$	—	Sleep staging
fills	$\alpha 1$ median (0.85), LF/HF 2.0, HF 0	—	Sleep staging
augmentation thresholds	0.68 / 0.63 / 0.68 / 0.72 / 0.40	—	Watch augmentation
RHR vitals penalty	10	points per SD	Vitals
RR band / penalty	1 / 15	br/min / points per br/min	Vitals
RR population window / below	12–18 / 90	br/min / points	Vitals
temperature bands	0.3 / 0.5 / 1.0 $\rightarrow$ 100/75/50/25	°C	Vitals

Appendix B: Pseudocode

```

function selectSource(S, I):
    if count(I) < 120: return count(S) >= 120 ? S : FAIL
    if count(S) < 120: return I
    pct = 100*|#I-#S| / max(#I,#S)
    if not (#I < #S and pct > 5): return I
    gaps = [(I[k-1].end, I[k].t) for k where I[k].t - I[k-1].end > I[k-1].rr + 2000]
    fill = [s in S where any gap contains (s.wall ?? s.t)]
    if gaps empty or fill empty: return S
    return mergeDedup(I, fill, tol=50ms)      // rebase fill.t := fill.wall

function mergeDedup(A, B, tol):
    i=j=0; out=[]
    while i<|A| or j<|B|:
        if j>=|B| or (i<|A| and A[i].t <= time(B[j])):
            out += A[i]; ta=A[i].t; i++
            while j<|B| and |time(B[j]) - ta| < tol: j++
        else:
            out += rebase(B[j]); tb=time(B[j]); j++
            while i<|A| and |A[i].t - tb| < tol: i++
    return out

function detectOnset(beats):
    if |beats| <= 300: return none
    W = [(mid.t, 60000/mean(validRR)) for 120-beat windows step 30 if >60 valid]
    if |W| <= 15: return none
    ceiling = max(W.hr) - 0.5*(max(W.hr)-min(W.hr))
    for j in 5 .. |W|-11:
        if mean(W[j-5..j-1].hr) - mean(W[j..j+9].hr) > 8 and mean(W[j..j+9].hr) <
ceiling:
        return min(W[j].t, 20 min)
    return none

function dateUndated(rec):
    hk = HealthKitSleep(last 24h, rec.beats)
    if hk.start: return hk.start - (detectOnset(rec.beats) ?? 0)
    return now - duration(rec.beats)

function mapWallToBeat(offset, P):
    if P[0].wall is none: return offset
    target = P[0].wall + offset
    k = first index with P[k].wall >= target
    return (k exists ? P[k].t : P[last].t) - P[0].t

function isBreak(p, q):      // q follows p
    if q.t - (p.t + p.rr) > p.rr + 2000: return true
    if p.wall and q.wall: return q.wall - p.wall > q.rr + p.rr + 2000
    return false

function maskedRMSSD(vals /*(origIndex, rr) not flagged, valid*/, kept, breaks):
    d = [vals[k].rr - vals[k-1].rr for k
        if kept[k] and kept[k-1] and vals[k].idx == vals[k-1].idx+1 and vals[k].idx not
in breaks]
    return d empty ? 0 : sqrt(mean(d^2))

function selectWindow(P, flags, sleepStart?, wake?, base?):

```

```

if |P| < 120: return none
s = max(0, map(sleepStart)) or 0
e = wake ? (map(wake) > s ? min(map(wake), dur) : dur) : dur
early = clampToRec(s + 0.3(e-s)); late = clampToRec(s + 0.7(e-s))
band = [first t>=early .. last t<=late]; B = |band|
size = B>=400 ? 400 : (B>=240 ? B/2 : max(60, 0.6B)); step = max(10, size/10)
C = scan(band, size, step, cap=0.10)
if C empty: C = scan(band, size, step, cap=0.15)
if C empty: return none
F = removeIsolatedSpikes(C, 1.5); cand = F empty ? C : F
O = [w in cand where organized(w)]
if O empty: return peakRMSSDFallback(...) // flagged not organized
if base: sort O by (Tier1(w.rmssd, w.meanHR, w.alpha1, base) desc, rmssd desc, pos
desc)
else: sort O by (rmssd desc, pos desc)
w = O[0]; w.consolidated = (w in F) and w.cv < 0.08
return w

function evalWindow(win, cap):
  if flaggedFraction(win) > cap: reject
  vals = unflagged valid beats; if |vals| < 50: reject
  kept = localGate(vals.rr, neighbours=10 excluding self, 20%)
  if count(kept) < min(300, max(50, 0.75|win|)): reject
  rmssd = maskedRMSSD(vals, kept, breaksIn(win))
  cv = popSD(keptRR)/mean(keptRR); alpha1 = count(kept) >= 64 ? DFA(keptRR) : none
  return {rmssd, cv, alpha1, meanHR, pos}

organized(w) = w.alpha1 ? (0.75 <= w.alpha1 <= 1.0 and w.cv < 0.08) : w.cv < 0.08

function admit(session, result):
  return session.overnight and quality not in {insufficient, preSleep}
  and result.windowLen >= 5 min
  and (result.organized or session.duration >= 3 h)

function baselineStats(points, scoredNightKey):
  P = [p where nightKey(p) < scoredNightKey] // leave-current-out
  if |P| < 3: return none
  R = last 60 of P; L = [ln p.rmssd for p in R if p.rmssd > 0]; if |L| < 3: return none
  w = |L| >= 7 ? 1 : sqrt(7/|L|)
  return {lnMean: mean(L), lnSD: max(sd(L)*w, 0.10),
    hrMean: mean(R.hr), hrSD: max(sd(R.hr), 0.5),
    cv7: |R|>=7 ? 100*cv(ln of last 7) : none, last: R.last.date}

function freeze(session, result, base, config):
  sleep, vitals = fetchWithTimeout(10s)
  vitals.restingHR = result.windowMeanHR ?? vitals.restingHR
  q = qualityDecision(result, sleep, base, session)
  inputs = q.useBaseline ? (exp(base.lnMean), base.hrMean, alpha1=none) : result
  bd = composite(inputs, sleep, vitals, base, ansBalance, referenceDate=session.end)
  if q.useBaseline and session.perceived: bd.HRV = 0.7*bd.HRV +
0.3*100*clamp(perceived)
  bd.version = "v3.1.oct2026"
  store(session.score = bd.composite/10, bd, frozenReadiness)

function autoSleepRefresh():
  s = most recent overnight with end within 18h, already scored
  if s.sleepUserAdjusted and s.snapshot exists: return
  f = HealthKitSleep(s); if not plausible(f, s): return

```

```

    upd = first or endLater>60s or more minutes or onsetMove>=15 or sourceUpgrade
    if not upd: return
    apply(f -> snapshot AND scoring boundaries)          // fill-only if user-adjusted
    if (first or |delta|>=20 or onsetMove>=15 or sourceUpgrade) and not userAdjusted:
rescore(s)

function stageNight(P, s, e):
    E = 5-min epochs in [s,e) with >=10 points and >=8 valid
    if |E| < 6: return none
    R = fractionalRanks(features(E), ties=mean, fill α1->median, lfhf->2, hf->0)
    for each epoch i: f = (mid_i - s)/(lastMid - s)
        deep, rem, awake = weighted sums per Section 3.9.3; if mid_i - s < 60 min: rem =
0
        stage_i = awake>0.80 ? AWAKE : deep>0.60 and deep>=rem ? DEEP : rem>0.63 and
rem>deep ? REM : CORE
        for i in 1..n-2: if stage_i != AWAKE and stage_{i-1} == stage_{i+1} != stage_i:
stage_i = stage_{i-1}
    return intervals(split on stage change or epoch gap)

```

Appendix C: Source Map

Method	Files (under Emuqu/Sources/)
Source selection, gap fill, de-duplication	Emuqu/Sources/Collection/DataSourceSelector.swift
Undated recording dating	Emuqu/Sources/Collection/RRCollector+DeviceRecording.swift, Emuqu/Sources/Analysis/SleepBoundaryResolver.swift
HR sleep estimate	Emuqu/Sources/Collection/HRSleepEstimator.swift, Emuqu/Sources/Services/MorningProcessingService+Sleep.swift
Append-only backup	Emuqu/Sources/Storage/RawRRBackup.swift, Emuqu/Sources/Storage/RawRRBackup+Streaming.swift
Rescue before clear	Emuqu/Sources/Collection/StrapStartSequence.swift
Pause/resume merge	Emuqu/Sources/Collection/RRCollector+PauseResume.swift, Emuqu/Sources/Collection/RRCollector+OvernightStreaming.swift
Artifact classifier	Emuqu/Sources/Analysis/ArtifactDetection.swift, Emuqu/Sources/Utilities/Constants.swift
Recording break, masked RMSSD (reporting)	Emuqu/Sources/Analysis/TimeDomainAnalysis.swift

Method	Files (under Emuqu/Sources/)
Window scan, evaluation, gates	Emuqu/Sources/Analysis/WindowSelection.swift, Emuqu/Sources/Analysis/WindowSelection+Evaluation.swift, Emuqu/Sources/Analysis/WindowSelection+Filters.swift, Emuqu/Sources/Analysis/WindowSelection+Scoring.swift, Emuqu/Sources/Analysis/ScoredRecoveryBlock.swift
DFA	Emuqu/Sources/Analysis/DFAAnalysis.swift
Pipeline entry	Emuqu/Sources/Analysis/HRVAnalysisPipeline.swift, Emuqu/Sources/Services/MorningProcessingService+Analysis.swift, Emuqu/Sources/Collection/RRCollector+Analysis.swift
Baseline	Emuqu/Sources/Analysis/BaselineTracker.swift, Emuqu/Sources/Models/SleepSchedule.swift, Emuqu/Sources/Utilities/Constants+SleepAndDisplay.swift
Recovery score	Emuqu/Sources/Analysis/RecoveryScoreCalculator.swift, Emuqu/Sources/Analysis/RecoveryScoreCalculator+Tiers.swift, Emuqu/Sources/Analysis/RecoveryScoreCalculator+Composite.swift, Emuqu/Sources/Analysis/ScoreDetailBuilder.swift, Emuqu/Sources/Analysis/HRVThresholds.swift, Emuqu/Sources/Utilities/Constants+RecoveryScore.swift, Emuqu/Sources/Utilities/Constants.swift (ScoringWeights)
Vitals	Emuqu/Sources/Analysis/VitalsScoring.swift, Emuqu/Sources/Models/RecoveryVitals.swift
Freeze and quality decision	Emuqu/Sources/Services/SessionAcceptanceService.swift
Version	Emuqu/Sources/Analysis/ScoringVersion.swift
Automatic refresh	Emuqu/Sources/Collection/SleepRefreshPolicy.swift, Emuqu/Sources/Collection/RRCollector+Reanalysis.swift
Cross-device merge	Emuqu/Sources/Storage/CloudKitSyncSupport.swift, Emuqu/Sources/Storage/CloudKitSessionFreshness.swift, Emuqu/Sources/Storage/SessionMerger.swift
Manual-window preservation	Emuqu/Sources/Services/ReanalysisService.swift
Sleep staging, Watch augmentation	Emuqu/Sources/Analysis/HRVSleepStageClassifier.swift, Emuqu/Sources/Analysis/HRVSleepStageClassifier+Watch.swift, Emuqu/Sources/Utilities/Constants+SleepAndDisplay.swift
Validation register / data	Tools/science_register/register.json, docs/hrv-reference-data.md